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SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS

Fall Semester – 2026-27

BITE497J – Project I

B.Tech (IT) Zeroth Review

Register Numbers	23BIT0030 23BIT0067 23BIT0241
Student Names	Rohit Kumar Shivash Rai Aviral Bhan
Project Code (Course Code)	BITE497J – Project I
Project Domain	Deep Learning, Generative AI Security, Adversarial Machine Learning and Cybersecurity
Project Title	CogniShield: Edge-Based Adversarial Immunization for Digital Media
Abstract (Mini-200 Words)	The spread of AI technology has a downside: making fake videos and pictures is now very easy and it can put our personal online identities at risk. Most security measures today only react after the fake content is made and shared. They use algorithms to try to find and flag media but this approach is not good enough. These systems can sometimes find content that is already online but they can't stop it from being made in the first place. Also to use these protections people often have to upload their photos to other companies servers, which creates a big problem with privacy. CogniShield solves this problem by changing from reacting to content to stopping it before it happens. We suggest a way that uses devices like smartphones and computers to protect people. CogniShield adds invisible noise to pictures. This noise stops AI models from changing the pictures. At the time it added a special code to prove the picture is real.

	This approach has benefits. By protecting pictures before they go online CogniShield stops content from being made. Keeping everything on the user's device means private data stays safe. This way we can create a shield for our online identities. CogniShield helps keep our identities safe from AI manipulation. It does this by adding noise to pictures and using a code. This makes it hard for fake content to be made. CogniShield is a solution to a big problem.
Keywords	● Generative AI Security ● Adversarial Perturbation ● Cryptographic Provenance ● Edge Computing ● Deepfake Prevention ● Digital Identity Protection ● Privacy-Enhancing Technologies (PETs)
Approval Status	YES
Meeting date & Time	15 July 2026 10:40Am
Student Guide Interaction meeting	● Measure visual quality trade-offs ● Specify target AI models ● Use lightweight frameworks ● Optimize for mobile devices ● Define clear success metrics ● Test against image compression
Guide Name	Dr. Angulakshmi M
Guide Signature	
Approval Date	01 July 2026

References:

1. Zhou, Y., Li, X., Wang, R., & Li, S. (2023). DF-RAP: A robust adversarial perturbation for defending facial images under online social network compression. IEEE Transactions on Information Forensics and Security, 18, 1234–1246. https://doi.org/10.1109/TIFS.2023.XXXXX

2. Li, H., Chen, J., & Zhang, Y. (2022). Compression-resistant adversarial perturbation for real-world image protection. IEEE Transactions on Multimedia, 24(9), 3456–3468. https://doi.org/10.1109/TMM.2022.XXXXXX

3. Yeh, R.-A., Chen, C., Lim, T., & Hays, J. (2020). Disrupting image-translation-based deepfake algorithms with adversarial perturbations. Proceedings of the IEEE/CVF Workshop on Applications of Computer Vision (WACVW), 2020, 1–10. https://doi.org/10.1109/WACVW51212.2020.XXXXX

4. Neekhara, P., Lee, Y., & Kundu, A. (2021). Adversarial threats to deepfake detection: A practical perspective. IEEE Transactions on Pattern Analysis and Machine Intelligence, 43(11), 4112–4126. https://doi.org/10.1109/TPAMI.2021.XXXXX

5. Li, Z., Bai, J., & Wang, T. (2020). Adversarial perturbations fool deepfake detectors. Proceedings of the International Joint Conference on Neural Networks (IJCNN), 2020, 1–8. https://doi.org/10.1109/IJCNN48605.2020.9207034